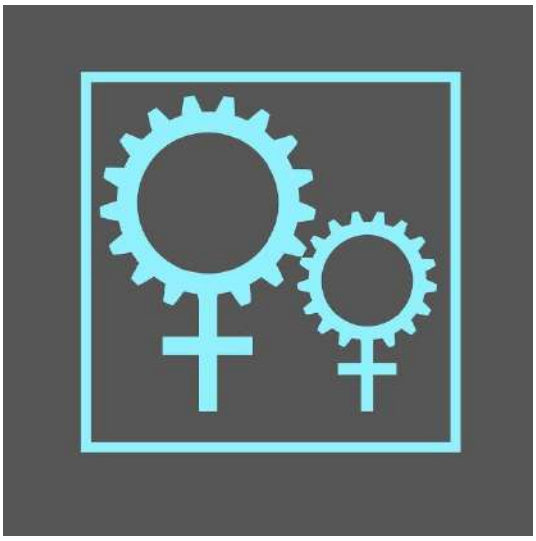


# Sisters of the Motherboard

# FTC Team #7444

# Engineering Portfolio

Note: References to supporting pages in the engineering notebook are made throughout the portfolio. Page numbers are given in terms of specific sections of the notebook. To view the referenced pages in the full notebook, refer to the "Summary Page" at the beginning of the full notebook where there are links directly to the corresponding pages.

# INTRODUCING OUR TEAM



## ABOUT OUR TEAM AND OUR SCHOOL

This is our eighth year of robotics and it has given us many interesting challenges. We have created unique ways to meet throughout Covid either via zoom or meeting in a large building and taking all of the CDC safety precautions. Years past have taught us about never underestimating the power of being proactive, that has come in handy this year especially with the limit on in-person meetings. However, we have learned many of the techniques to work around the physical barriers, like our Tuesday class time where we meet over zoom and our Thursday time where we are fortunate enough to be able to work in person. Along the way, we have made it our personal goal to persevere through the tough times, to make sure we stay connected, and to stay a team.

## HOW OUR TEAM WORKS/TEAM STRUCTURE

Sisters of the Motherboard has multiple different subgroups to specialize in areas team members are interested in. These subgroups include building, programming, CAD/3-D printing, outreach, engineering notebook, social media, and website design. Most team members dabble in multiple subgroups especially in the outreach, engineering notebook, social media, and website design subgroups. We have traditionally had a small group of two or three seniors as the team leaders. This year since we have a smaller team due to Covid we have a smaller team and therefore a smaller group of leaders. Helen and Mandy have taken the leadership role and have been working very hard to keep the team running steadily and effectively.

# TEAM PLAN

## ORGANIZATION OF THE SEASON

This year we had to adjust the organization of the season from previous years. We were more lenient on deadlines and specific goals we wanted to accomplish because of Covid. I think this allowed us to focus more on the work and the robot than the deadlines. That being said we have held a loose structure for our robotics class this year. We meet every Tuesday over zoom, every Thursday at Kaleideum, and occasionally we have Saturday build sessions. We also have worked to keep the engineering notebook on a schedule so that we finish all we can before things get crazy during competitions.

## IMPROVEMENTS FOR NEXT SEASON

Covid permitting, we are going to try to resume classes which would allow us to have a long double period on either Tuesdays or Thursdays. A double period allows us to work uninterrupted for a longer period which is more beneficial than meeting a few times for shorter intervals. Next season we are also working on a more structured build and coding schedule to keep us on track.

## MENTORING STRUCTURE

We pride ourselves on our mentoring structure, which consists of older members teaching younger members different subgroups of robotics they are interested in. We have this because we think it is very important to have the people you are going to be working with show you how to do something, whilst you get a feel of how it would be to work with that person. Older members make sure they pass down all the knowledge they can because once they graduate and leave it is up to the members they trained to lead the team and pass down their information. I think we are definitely more open to using outside material, like videos from other teams this year because it is so hard to meet in person and pass knowledge like we normally would. We are currently working with a structure where older team members work on the robot and the younger members join in and ask questions when they are confused. We think this approach helps because the older members are showing the younger members how to code, build, CAD, or whatever else they are interested in while still making progress on the robot. Our goal is to have our younger members

confident enough to show the even younger members different skills by the end of their first year. We also like to collaborate with our sister team #15333 Code Sisters because it gives us more people to bounce ideas off of and we both benefit from mentoring each other when we need help.

| BUILDING | CAD | PROGRAMMING | DRIVE TEAM | SOCIAL MEDIA ENGINEERING NB |
|---|---|---|---|---|
| **Emma** | **Margot** | **Helen** | **Mandy** | **MB** |
| **Claire** | **Andreanna** | **Mandy** | **Emma** | **Symone** |
| **Anne** | **Lily** | **Emma** | **Claire** | |

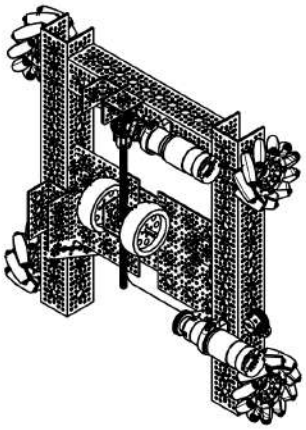
## FINANCIAL STABILITY

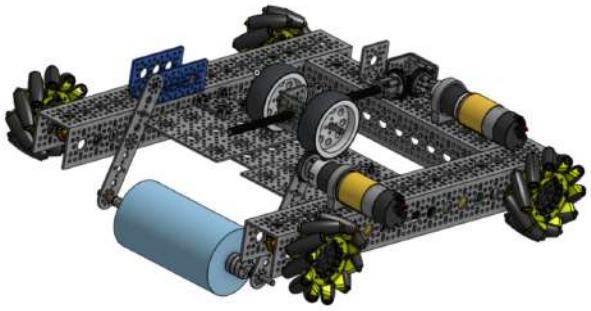
Since we were not able to host camps this year we were not able to get our main source of funding. All of our funding sources came from donations of alums and a donation from one of the people we are connected with. In the future we can definitely improve on making alum donations easier and better announced to the Salem community. We can also improve on looking for grants, connecting with alums, partnering with local businesses for sponsorships, and fundraising if allowed by the school. We are taking steps toward writing grants now to support our financial stability and will continue to do so in the future.

# ABOUT OUR ROBOT

## OVERALL DESIGN

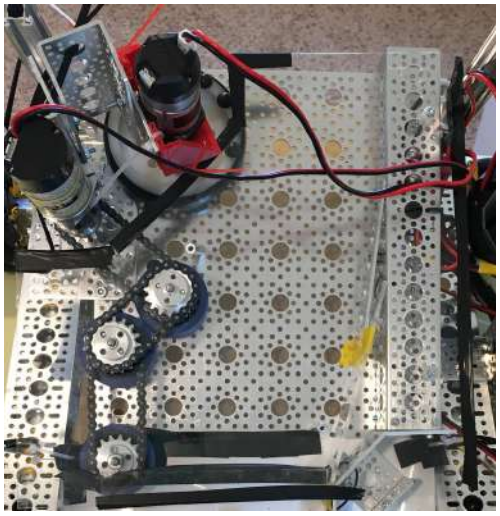
Our robot, Gerty, has a goBilda strafer chassis with a wobble goal grabber and three main systems, an intake, outtake, and lead screw mechanism.

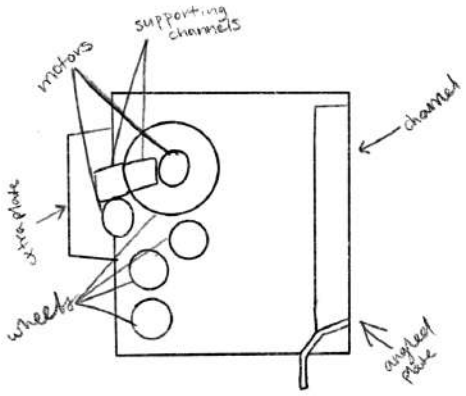




## LAUNCH MECHANISM

- Our launch mechanism is based on a fly-wheel outtake, with one main wheel and three smaller wheels leading up to it to help guide the ring up the ramp
- **See Engineering Notebook Design Section pages 16 and 19 for more Outtake mechanism design process.**

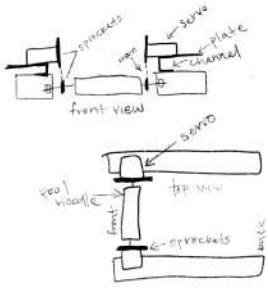


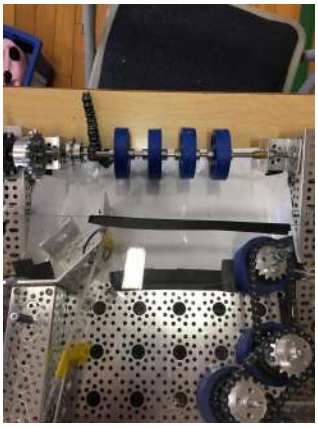


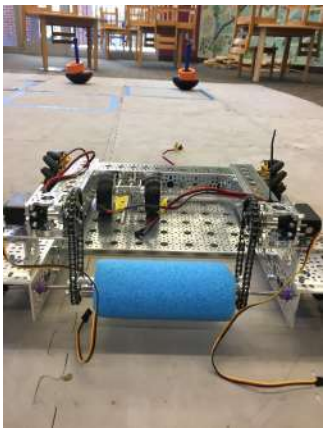
## INTAKE

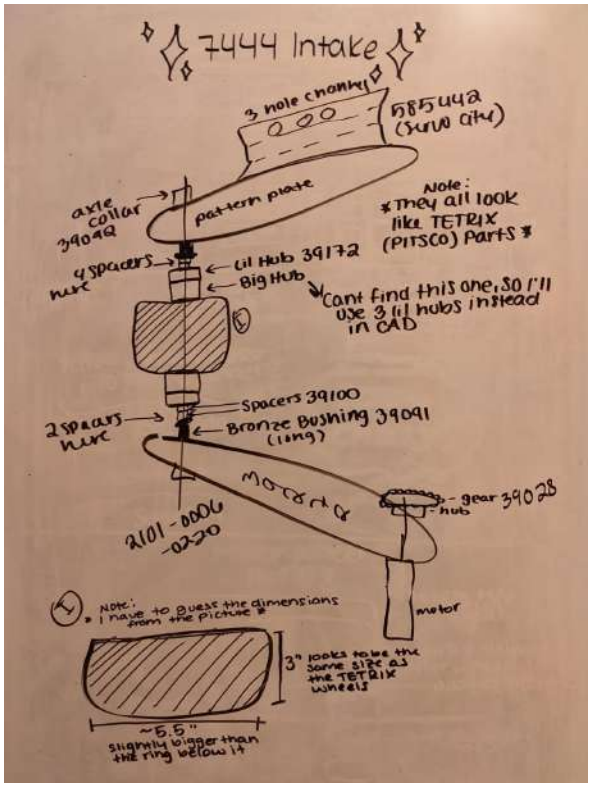
- Our intake system is pasta roller-inspired, using chains and servos for rotation and a pool noodle and air bag material for grip
- The intake is currently in a state of transition as we are developing a new design to help pull the ring on to the ramp of our outtake

**See the Engineering Notebook Design Section page 14 for more on the Intake mechanism design process.**



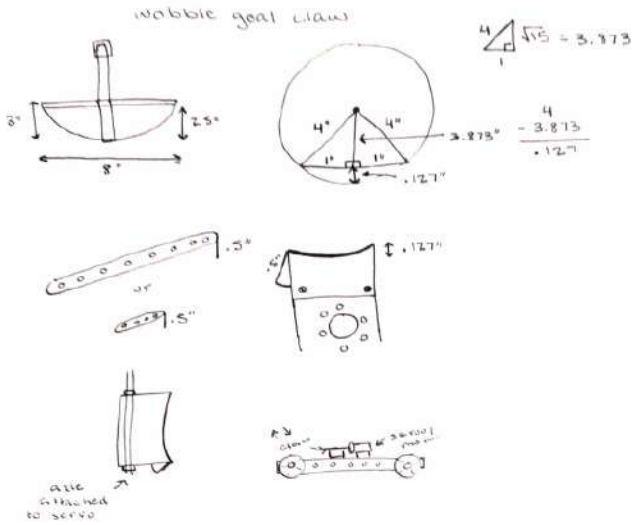








## WOBBLE GOAL CLAW DESIGN

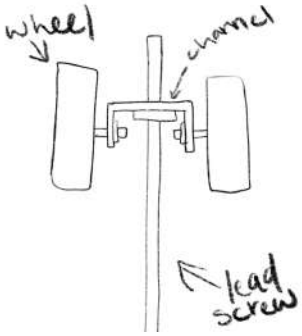
- Wobble goal claw design
- Pythagorean theorem used to calculate the distance between the corners and midpoint of curve
- Designed to be 3-D printed



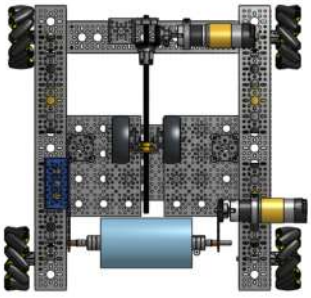
## LEAD SCREW MECHANISM

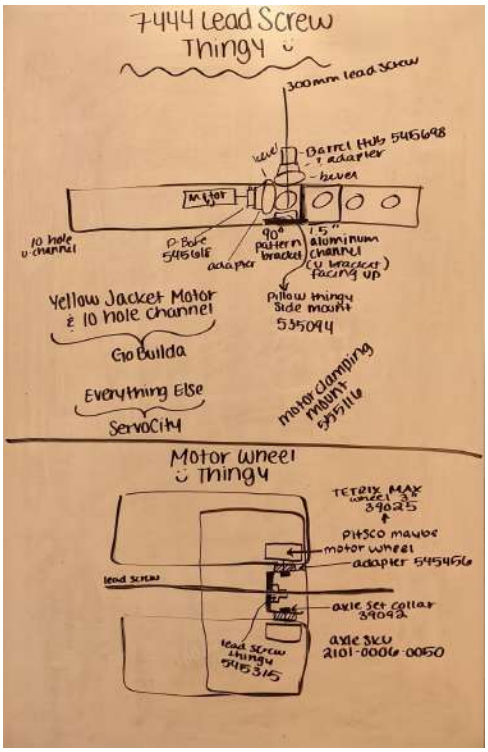
- In order to control the angles of our outtake, we created a lead screw system with two wheels that roll along the base of our outtake, raising it up or lowering it depending on which direction the wheels roll
- This idea came from a similar system on last year's robot that used a system of chains and bevel gears to lower and raise a platform
- Our original design included goRails and extra channels for stability, though oru current iteration was designed to save as much space under the outtake as possible so it could be used for battery packs, wires, etc.
- Currently, our lead screw has 3 set positions, one for each angle needed to launch the rings into the goal from the start line. There is also a manual option on our gamepad to adjust the angles as needed
- **To see more specifics of our lead screw development process, reference pages 14 and 52 in the Weekly Meetings section and pages 4 and 6 in our Design Section of the Engineering Notebook.**









# ABOUT OUR DESIGN PROCESS

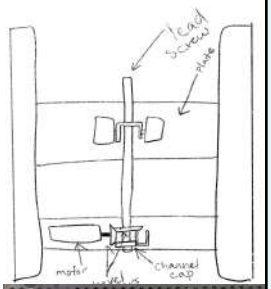
## BUILDING

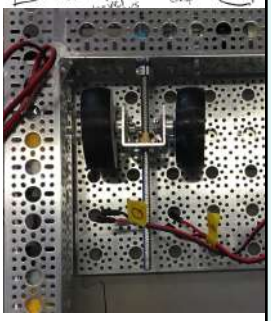
- In the beginning due to COVID restrictions, we were not able to access the robot for many months, which created issues for our testing and prototyping timeline
- We were not able to meet in person until early November and only began the construction of our robot a couple of weeks ago, however we were able to send the robot home with team members to continue building between meetings to make up for the time lost in the fall
- Since our timeline and plans were so altered during this time, we have had to build the robot in tight time constraints, which, has helped our team grow in knowledge of how our robot works and understand the importance of working together to finish the job



**Challenges in Design/Build**

- One challenge we had was the placement of our lead screw on our chassis
- Originally we had the screw on top of the main crossbar, but that raised the outtake system to far for the intake system to pull the ring up to it, and was not very stable
- We moved the lead screw so the motor was inside the cross bar to lower it and so we could thread the lead screw through the channel for stability, but we discovered the bevel gears we were using were to large to properly engage in that space
- As a result, we switched to using a 2:1 gear system inside the cross bar to allow for that space though still had trouble working in such a tight space with the gears
- **To see more specifics of our lead screw development process, reference pages 14 and 52 in the Weekly Meetings section and pages 4 and 6 in our Design Section of the Engineering Notebook.**



## CODING

- We faced challenges with bring in a virtual environment and not being able to access the robot to test, so we utilized a virtual robot emulator to test autonomous code
- We also used Github to upload updated versions of our code to help those who are remote keep up-to-date with adjustments

**Challenges with Code**

- A main challenge was working with encoders to run our lead screw system
- We hoped originally to use encoders to set “levels” or angles for the lead screw, each angle corresponding to a button on the gamepad
- Currently, we are working to create conscience, effective code to achieve this but for now we are achieving the same effect using if/then statements to run the lead screw for a certain number of seconds depending on the button pressed

# ABOUT OUR GAME STRATEGY

## OVERALL APPROACH TO THE GAME

Our overall approach to the game is finding consistent scoring opportunities we can do well. Our design process has been about perfecting our intake and outake system and so our main scoring plan is through the rings. We also plan to push the wobble goal and park in both autonomous and end game.

## AUTONOMOUS

Our Autonomous code works to park on the launch line from every starting location but our main focus will be depositing the wobble goal in one of the three target zones. We have utilized pre-programmed drive and turn methods constructed in the hardware class to simplify our autonomous code and for more precise turns. Additionally, we have implemented telemetry to receive information about the robot's status. We have paths for each target zone and a path to park after delivery. Our focus on simple yet consistent paths will help us always optimize our autonomous points. We have greatly increased our efficiency by problem-solving a method to connect the control hub via bluetooth to our computers, allowing for wireless code download **(see more on page 54 of the Weekly Meetings section of the Engineering Notebook and page 5 of the Programming section of the Engineering Notebook).**

## DRIVER-CONTROLLED

Our main goal for the driver-controlled period is to score as many rings as possible in the goal slots. We have pre-programmed ramp heights that allow us to shoot into all three goal slots accurately for ease during game play.

## END GAME

Our endgame strategy will be to knock down the power shots, push the wobble goal to the designated position, and park in specific locations.

# CONNECTING WITH THE COMMUNITY

## UNIQUE CHALLENGES OF OUTREACH IN 2020-2021

Outreach is typically a large component of our robotics teams, with several summer STEAM camps and volunteer experiences throughout the year. We truly believe that creating the robot is only part of the overall mission of a team, and emphasize spreading the FIRST message and its core values of empowering young people and building relevant skills through STEM education. However, due to the unique constraints that have come with the Covid-19 pandemic, we have not been able to hold our usual camps and take advantage of as many volunteer opportunities. While we are limited during this time, we are also proactively seeking to prepare for future outreach by making new community connections, including setting up a separate workspace at Kaleideum. Over Jan Term, 2 members of #7444 and 1 member of #15333 volunteered as counselors for a kids' science camp held here, where they were able to brainstorm and test some new activities that may be integrated in later STEAM camps. Team members have been working diligently over Jan Term to plan more activities for future camps as well. **See page 6 of Kaleidium Outreach in our Engineering Notebook for more information.**

Another difficulty we have overcome this year as a team has been communicating and collaborating effectively throughout an entirely remote fall semester. Since we are a boarding school, team members were dispersed throughout parts of the US and even the world as they studied from home, meaning we have also contended with varied time zones and limitations to those who can physically build/program the robot. We have continued to hold weekly meetings for both teams, utilizing breakout rooms to work in subgroups. Additionally, we have met in small groups at our Kaleideum workspace roughly once per week, following appropriate guidelines and zooming in those unable to attend in-person. While this year has undoubtedly brought novel challenges for all, the ways in which we have creatively sought to overcome these obstacles has strengthened our bond as teams.

## CONNECTING WITH THE STEM AND FIRST COMMUNITY

### MOUNTS ROBOTICS CENTER + KALEIDEUM

**About**

- Mounts Robotics Center has been a partner of our team for the past few years and is a local organization surrounded around all things STEM and helping students get engaged. This year, they have worked with us to partner with Kaliedium, a local children's science museum.

**Relationship**

- For the past few years we have been partners with Mounts Robotics Center, utilizing their mentors and materials. From Kaliedium, we have been able to use a field space in their museum.

**Goals and Outcomes**

- Our goals with Mounts is to contribute to the STEM community of Winston-Salem and firmly cement robotics as an important part of our community. Kaliedium has given us the opportunity to work with kids at the museum to expose them to robotics and help get them excited about robotics and STEM overall.

### FIRST LADIES

**About**

- First Ladies is a FIRST organization of girls robotics teams that encourages young women to explore the STEM field, hosts events to showcase current women in STEM, and runs a blog about what it means to be a woman in STEM.

**Relationship**

- We are a proud member of First Ladies and as such we contribute to their blog and host FIRST Ladies events.

**Goals and Outcomes**

- Our goals are to unite women in a male dominated STEM field and we have succeeded in that goal at Salem Academy, meanwhile we are trying to spread that unity to the community around us.

## FIRST ROBOTICS TEAM: GIRLS ON FIRE (FRC #5679)

**About**

- Girls On Fire (GOF) is one of two all-girls FRC teams in North Carolina and the only all-girls team in the Triad. They empower girls to pursue STEAM and develop life-long skills such as communication, teamwork, business, and design, as well as mechanical, programming, and electrical skills used with the robot.

**Relationship**

- GOF and Sisters of the Motherboard have been connected for a couple of years, sharing mentors and team members over the years. Currently, two Salem students are a part of GOF, Claire from #7444 and Michaela from #15333. Mentors from GOF are also a part of MRC, allowing us to use space at Kaleideum when we gather in-person.

**Goals and Outcomes**

- In partnership with GOF we hope to spread awareness and recruit more young women into FIRST, and provide an environment where they can thrive in STEAM. GOF has helped provide feedback and solutions to challenges for our robot and helped us connect with other resources, such as Kaleideum and MRC. Partnering with GOF also gives Salem students experience and access to different awards and scholarships that can benefit them as they head off to college and into their careers.

## WAKE FOREST UNIVERSITY COMPUTER SCIENCE DEPARTMENT

**About**

- The Wake Forest University Computer Science and Math Departments, home to FIRST alumni and students currently conducting computer science research, has been a partner of ours this season.

**Relationship**

- Through this partnership we have had access to mentors who have taught us programming skills and assisted us in our coding for this year's challenge.

**Goals and Outcomes**

- We are working with current students at Wake to start a Robotics club that will be a partner of ours and allow us a closer relationship with a local university, professional mentors, and resources from the Wake Forest makerspaces. We hope this club at Wake will be a lasting partnership in the coming years.

# MOTIVATING OTHERS TO BECOME INVOLVED IN STEM

## KALEIDEUM JANUARY TERM INTERNSHIPS

**About**

- As our opportunities for outreach- notably STEAM camps we would normally organize- have been limited, we were incredibly excited to have 3 robotics team members volunteer as counselors at a science camp at Kaleideum this year. Mandy and Helen from #7444 and Claire from #15333 assisted with organizing STEM-oriented camp activities and engaged the k-5 age campers as volunteers throughout a 2-week period.

**Relationship**

- One of our mentors from MRC who has also worked with us in securing a new building space at Kaleideum graciously agreed to facilitate this internship, and one of our team mentors helped us coordinate plans for activities with Kaleideum staff. We met with the camp coordinator to discuss plans and what responsibilities we would hold as volunteers the week before winter break, and continued working with staff leading up to our internship.

**Goals and Outcomes**

- This camp was a great opportunity to brainstorm and implement some activities we may be able to incorporate into our future STEAM camps. Volunteers were challenged to create new activities for afternoon stations. Upon completing the internship, they assessed in detail the successes and shortcomings of each for future reference. In addition to expanding our arsenal of activities, the camp has allowed an opportunity for volunteers to have fun in working with aspiring scientists!

## GIRL SCOUTS - ROBOTICS AND CODING BADGES

**About**

- Salem Academy has been partnering with the Girl Scout to create camps that allow Girl Scouts to earn various robotics badges. Over Janterm our team has been creating day camps with activities that can be modified to become either virtual or in person! **To see more, reference page 28 in the Weekly Meetings section of the Engineering Notebook.**

**Relationship**

- Sisters of the Motherboard has done events in the past with the Girl Scouts, but this is the first time we are looking to help them earn badges. We create different camps and activities that Girl Scouts can do to earn

badges and learn about STEM. As an all girls school supporting an all girls organization is very important to Salem Academy.

**Goals and Outcomes**

- Our goal is to host a day camp in the Spring 2021 to allow troops to earn their Junior Coding and Robotics badges. We are also hoping to encourage young girls to be interested in and pursue the STEM field through robotics. It goes back to our mission statement about wanting to encourage the passion of young girls in a safe and fostering environment.

## STEAM SUMMER CAMPS

**About**

- In a typical year, our robotics teams host 2 separate, week-long summer STEAM (A for arts) camps for kids. One is co-ed at Hatteras, and the other all-girls here at Salem. During these camps, robotics team members organize and facilitate STEM-oriented activities and engage with campers. We were unfortunately unable to hold these camps this year given the pandemic, however we look forward to resuming them as soon as we can!

**Significance**

- The summer STEAM camps mean a lot to our robotics program, being an opportunity to interact with these bright young people and promote values of FIRST. It has been extremely gratifying to see some campers return year after year, excited for the opportunities to program EV3s, observe self-made slides through foldscopes, make slime, and participate in countless other fun activities. Additionally, the camp at Salem has allowed us to channel our mission of exposing young girls to robotics and STEM opportunities in general, as providing a supportive space for girls to explore STEM is one of our integral values as an all-girls team.

**Goals and Outcomes**

- Through participating in the STEAM camps, team members gain invaluable experience working with kids and learn to curate activities that will prove most engaging to a variety of age groups. Many of the activities we incorporate each year are staples that have proven successful and been reused, however we try to integrate several new ideas at each camp to improve these based on assessment of their success in the future. In addition to building this practical skill, participating in these STEAM camps is a gratifying experience, motivating us to continue improving and standing for our values in our own endeavors.

## VIRTUAL SCIENCE WORKSHOPS

**About**

- Some of our robotics girls came up with STEM ideas for kids to do over zoom. These are short activities that can be done a multitude of times and still be fun and engaging for the kids, as well as being able to do it remote or in person. They also include doing robot demonstrations with the finished robots after the end of the season.

**Significance**

- This is a fun way to teach kids about STEM and get them to interact with others, even though it's virtual. We have typically done day long events, so having some shorter sessions opens up opportunities for the types of activities we can do.

**Goals and Outcomes**

- The goal is to teach kids about STEM careers, and help them develop an interest in those fields. Kids love seeing the "big kids" and role models are very important at this age.

## NEWSLETTERS AND SOCIAL MEDIA




**About**

- Newsletters are sent out every week describing the activities that our robotics girls did that week. There are also social media platforms (Instagram Twitter, Facebook) that are used to show the events we go to and the overall fun of our robotics team!

**Significance**

- We are able to communicate with parents and the other girls of the school, showcasing the amazing things robotics does. It is an effective and efficient way to communicate to the rest of the team as well. This is especially significant given the times that we are in.

**Goals and Outcomes**

- The goal is to be able to efficiently communicate with the other members of the team, as well as members of Salem Academy, so everyone can be up to date on the robotics team's matters.
- Connect with other FIRST teams to provide support (last year we helped another FTC get their Instagram started!)
- Connect with STEM professionals and companies for possible mentorship and sponsorship.